Suggested problems: 1-51 odd

Problem 1: Compute the following iterated integrals.

(a) $\int_0^3 \int_0^4 (4x + 3y)\ dx\ dy$

$$\int_0^3 [2x^2 + 2xy]_0^4\ dy = \int_0^3 (32 + 12y)\ dy = 32y + 6y^2\Big|_0^3 = 150.$$

(b) $\int_0^2 \int_0^3 (x^2 + y^2)\ dy\ dx$

$$\int_0^2 \left[x^2 y + \frac{y^3}{3}\right]_0^3\ dx = \int_0^2 (3x^2 + 9)\ dx = x^3 + 9x\Big|_0^2 = 26.$$

(c) $\int_0^1 \int_0^1 ye^{xy}\ dx\ dy$

$$\int_0^1 e^{xy}\Big|_0^1\ dy = \int_0^1 (e^y - 1)\ dy = e^y - y\Big|_0^1 = e - 2.$$

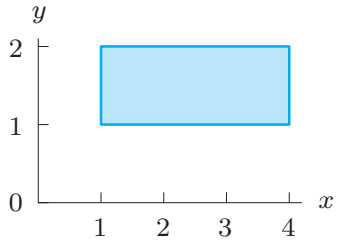
Problem 2: A building is 8 meters wide and 16 meters long. It has a flat roof that is 12 meters high at one corner, and 10 meters high at each of the adjacent corners. What is the volume of the building?

See example 1 on page 876 of the text:

$$\int_0^{16} \int_0^8 \left(12 - \frac{1}{4}x - \frac{1}{8}y\right)\ dx\ dy = 1280.$$

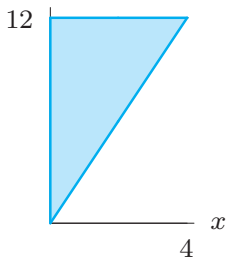
Problem 3: For each part below, write $\int_R f\ dA$ as an iterated integral of the shaded region R.

(a)



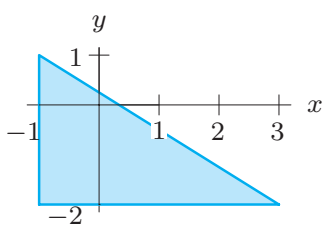
$\int_1^2 \int_1^4 f(x,y)\ dxdy$

(b)



$\int_0^{12} \int_0^y f(x,y)\ dxdy = \int_0^4 \int_x^{12} f(x,y)\ dydx$

(c)



$\int_{-2}^1 \int_{-1}^{-\frac{4}{3}y+\frac{1}{3}} f(x,y)\ dxdy = \int_{-1}^3 \int_{-2}^{-\frac{3}{4}x+\frac{1}{4}} f(x,y)\ dydx$

Problem 4: Compute the following iterated integrals. Sketch the region R of integration.

(a) $\displaystyle\int_0^2 \int_0^x e^{x^2}\ dy\ dx$

$$\int_0^2 \int_0^x e^{x^2}\ dy\ dx = \int_0^2 e^{x^2} y\Big|_0^x\ dy = \int_0^2 xe^{x^2}\ dx = \frac{1}{2}e^{x^2}\Big|_0^2 = \frac{1}{2}e^4 - \frac{1}{2}.$$

(b) $\displaystyle\int_1^4 \int_{\sqrt{y}}^y x^2y^3\ dx\ dy$

$$\int_1^4 \int_{\sqrt{y}}^y x^2y^3\ dx\ dy = \int_1^4 \frac{1}{3}x^3y^3\Big|_{\sqrt{y}}^y\ dy = \frac{1}{3}\int_1^4 (y^6 - y^{9/2})\ dy = \frac{1}{3}\left[\frac{y^7}{7} - \frac{2}{11}y^{11/2}\right]_1^4 = \frac{151555}{231}.$$

(c) $\displaystyle\int_1^5 \int_x^{2x} \sin(x)\ dy\ dx$

$$\int_1^5 \int_x^{2x} \sin(x)\ dy\ dx = \int_1^5 y\sin(x)\Big|_x^{2x}\ dx = \int_1^5 x\sin(x)\ dx = [-x\cos(x) + \sin(x)]_1^5 \approx -2.68.$$

Problem 5: Compute the following iterated integrals by reversing the order of integration.

(a) $\int_0^2 \int_y^2 e^{x^2}\ dx\ dy$

$$\int_0^2 \int_0^x e^{x^2}\ dy\ dx = \frac{1}{2}e^4 - \frac{1}{2}.$$

(b) $\int_0^1 \int_{x^2}^1 \sqrt{y}\sin(y)\ dydx$

$$\begin{aligned}\int_0^1 \int_0^{\sqrt{y}} \sqrt{y}\sin(y)\ dxdy &= \int_0^1 x\sqrt{y}\sin(y)\Big|_0^{\sqrt{y}}\ dxdy \\ &= \int_0^1 y\sin(y)\ dy \\ &= [-y\cos(y) + \sin(y)]_0^1 \\ &= -\cos(1) + \sin(1).\end{aligned}$$